Business Performance Diagnosis Report

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Tech Stack: Python, Power BI

DATA AUDIT PROFILE:
Dataset Scope: 9,994 Retail Transactions (2014–2017) | Kaggle Dataset

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1. Executive Summary

Conducted a financial diagnostic on 9,994 retail transactions ($2.30M sales) to resolve chronic margin compression. Despite a 46.88% revenue surge (2014–2017), net margins declined to 12.74% in 2017. Root-cause analysis isolated aggressive discounting (r = -0.22) as the key profit leak, driving severe losses in Furniture (Tables losing -$17.7K) and dragging Central region margins to 7.92% (vs. West's 14.94%).

Core Metric / Dimension	Observed Baseline Value
Total Sales Revenue	**$2,297,200.86 (~$2.30M)**
Cumulative Net Profit	**$286,397.02 (~$286.40K)**
Overall Profit Margin	**12.47%**
Total Fulfilled Orders & Units	**5,009 Unique Orders | 37,873 Units Sold**
Cumulative 4-Year Sales Growth	**46.88% (2014 to 2017 Expansion)**
Most Profitable Category	**Technology (17.40% Margin | $145.45K Profit)**
Primary Profit Dilution Drag	**Furniture (2.49% Margin) & Severe Heavy Tables (-$17.7K)**

2. Business Problem Statement & Analytical Objectives

Why is the company's profit margin declining despite robust revenue expansion, and what three specific, data-driven actions should management implement to protect margins and restore profitability?

Core Analytical Deliverables & Scope

- Identify whether top-line revenue growth is accompanied by declining or compressing net profit margins across annual cycles.
- Determine the primary structural factors contributing to poor profitability (discounts, product mix, shipping, or regional overhead).
- Develop an interactive, multi-dimensional Microsoft Power BI dashboard to monitor enterprise KPIs, customer segments, and regional trends.
- Deliver three concrete, highly actionable, and mathematically supported recommendations to immediately improve profitability.

3. Dataset Architecture & Data Preparation Pipeline

The underlying dataset comprises 9,994 order line-items spanning 21 commercial features. To ensure data integrity, a rigorous Python (Pandas/NumPy) ETL pipeline was executed prior to analytical modeling and Power BI ingestion.

3.1 Data Cleaning & Feature Engineering in Python

1. Schema Standardization: Stripped trailing whitespaces, converted column headers to lower-case snake_case, and formatted date strings.
2. Data Integrity Verification: Validated 0 duplicate records and 0 null/missing values across all 9,994 transactions.
3. Date Casting & Temporal Features: Cast 'order_date' and 'ship_date' to datetime format, extracting 'order_year', 'order_month', and 'order_month_name'.
4. Operational Fulfillment Metrics: Engineered 'shipping_days' (ship_date - order_date) to monitor carrier transit latency.
5. Margin Ratio Computation: Calculated 'profit_margin' per line-item ((profit / sales) * 100) handling zero-division safeguards.
6. Clean Dataset Serialization: Exported validated structured records to 'superstore_cleaned.csv' for seamless Power BI data modeling.

3.2 Statistical Profile & Exploratory Audit

Pipeline Stage	Technology / Tools	Core Functionality & Deliverable
Data Ingestion & Cleaning	Python (Pandas, NumPy)	Header normalization, zero duplicate/null verification, Latin-1 decoding
Feature Engineering	Python (DateTime, Logic)	order_year, order_month, shipping_days, profit_margin_pct
Visual & BI Intelligence	Microsoft Power BI Desktop	DAX KPIs, interactive cross-filters across Region, Category & Year

4. Key Performance Indicators (KPIs) & Business Baseline

The enterprise KPI card row in the Power BI dashboard establishes the foundational financial health of the business:

2.30M Total Sales Revenue ($)	**12.47%** Overall Profit Margin (%)	**46.88%** Sales Growth % (4-Yr)	**286.40K** Total Net Profit ($)
5.01K Total Unique Orders	**37.87K** Total Units Sold	**15.62%** Average Discount Applied	**West** Top Profitable Region ($108.4K)

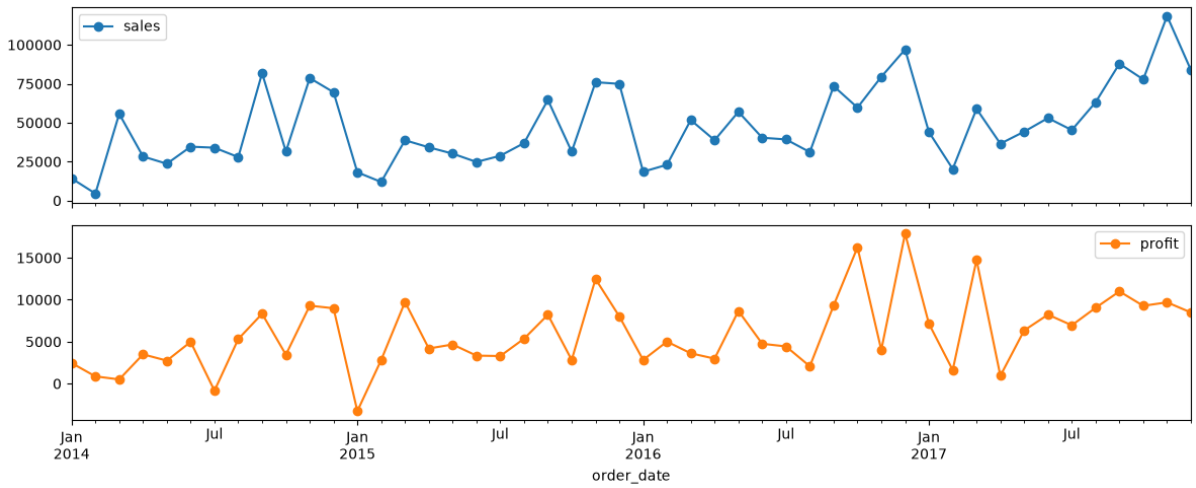
• Top-Line Revenue Scale ($2.30M)	The organization has attained significant commercial distribution scale across North America, transacting over 37,800 units across 5,000+ customer orders.

• Moderate Margin Efficiency (12.47%)	While an overall margin of 12.47% is positive, peer benchmark commercial distributors maintain 15%–18% operating margins, signaling systematic profit erosion.

• Robust 4-Year Expansion (46.88%)	Between 2014 and 2017, gross revenue surged by nearly 47%, yet net profit conversion did not scale in tandem due to heavy price promotions.

5. Year-over-Year Growth vs. Profit Margin Compression

To directly answer the core business question, yearly performance metrics were aggregated to track sales growth against net margin percentage. The data confirms that margin compression commenced in 2017 despite record revenue:



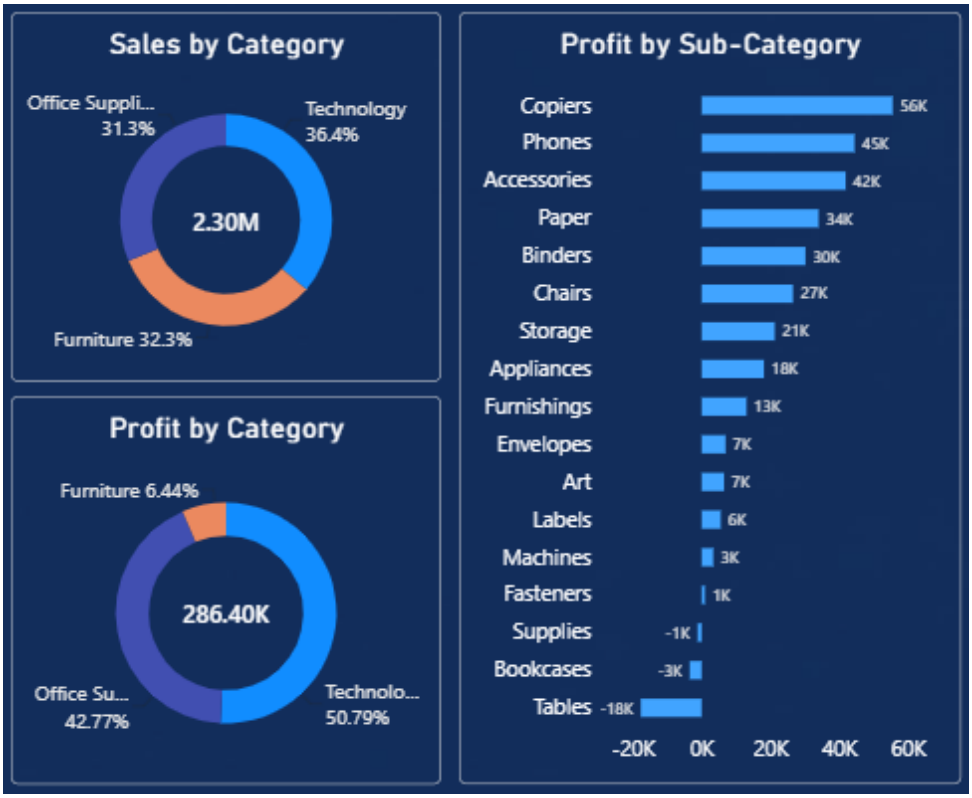
Year	Sales ($)	Profit ($)	Profit Margin (%)	Sales Growth (%)	Margin Change (pp)
2014	$484,247.50	$49,543.97	10.23%	Baseline	Baseline
2015	$470,532.51	$61,618.60	13.10%	- 2.83%	+ 2.86 pp
2016	$609,205.60	$81,795.17	13.43%	+ 29.47%	+ 0.33 pp (Peak)
2017	$733,215.26	$93,439.27	12.74%	+ 20.36%	- 0.68 pp (Compression)

• Empirical Proof of Margin Erosion	In 2017, sales surged by 20.36% (an incremental $124K in revenue), but profit margin declined by 0.68 percentage points (from 13.43% down to 12.74%). The enterprise is pursuing 'unprofitable growth'—selling more volume at structurally thinner margins.

6. Deep-Dive Profitability Breakdown

6.1 Category & Sub-Category Margin Disparities

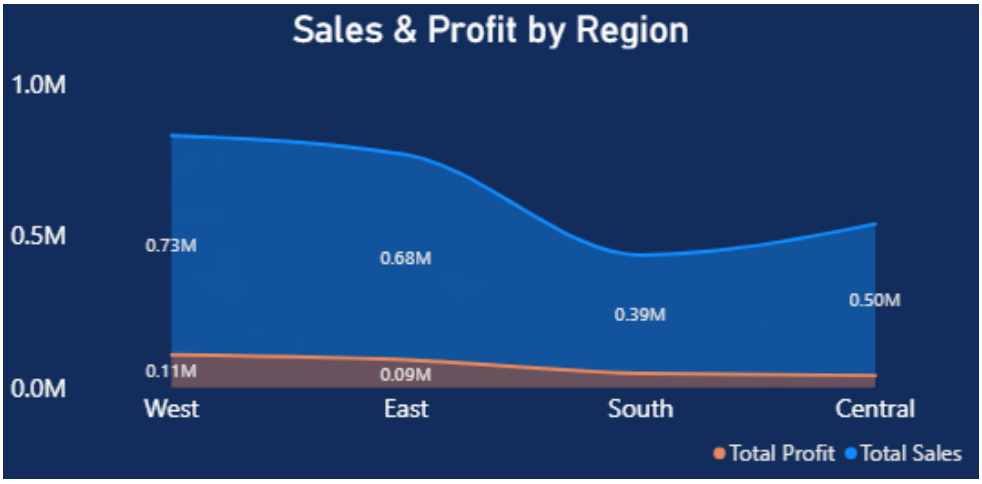
A major driver of margin dilution is product category imbalance. Technology and Office Supplies generate healthy margins (17.40% and 17.04% respectively), while Furniture operates at an alarming 2.49% net margin.



Category	Sales ($)	Profit ($)	Units Sold	Profit Margin (%)
Technology	$836,154.03 (36.4%)	$145,454.95	6,939	17.40% (Top Margin Driver)
Office Supplies	$719,047.03 (31.3%)	$122,490.80	22,906	17.04% (Volume Anchor)
Furniture	$741,999.80 (32.3%)	$18,451.27	8,028	2.49% (Critical Drag)

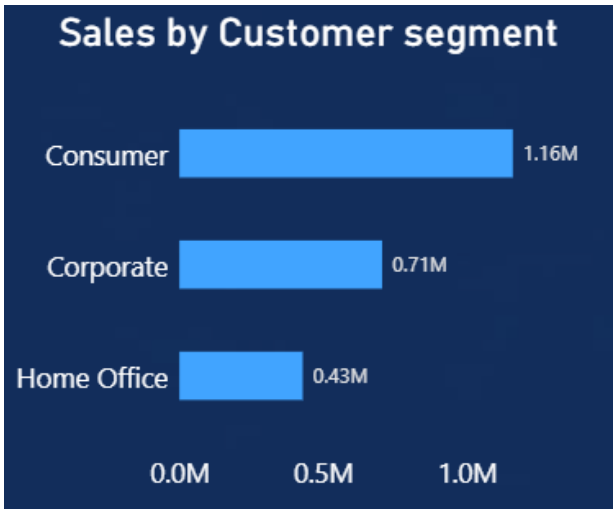
6.2 Regional Performance & Central Region Drag

Geographic cross-tabulation isolates a massive margin divide between the Western and Central operating divisions:



Region	Sales Revenue ($)	Net Profit ($)	Unique Orders	Profit Margin (%)
West	$725,457.82	$108,418.45	1,611	14.94% (Top Regional Margin)
East	$678,781.24	$91,522.78	1,401	13.48% (Healthy Operational Return)
South	$391,721.90	$46,749.43	822	11.93% (Stable Baseline)
Central	$501,239.89	$39,706.36	1,175	7.92% (Severe Operational Drag)

6.3 Customer Segment Contribution



7. Root Cause Analysis: The Discounting Trap

7.1 Statistical Correlation & Negative Profit Sensitivity

Python correlation matrix analysis exposes an explicit negative correlation between discount rate and net profit (r = -0.22). While sales moderately correlates with profit (r = +0.48), discounting directly erodes transaction margins. The scatter plot proves that orders discounted beyond 20% consistently drop below the zero-profit line into severe deficit.

Feature	Discount	Sales	Profit
Discount	1.00	- 0.03	- 0.22 (Negative Drag)
Sales	- 0.03	1.00	+ 0.48 (Volume Scale)
Profit	- 0.22	+ 0.48	1.00

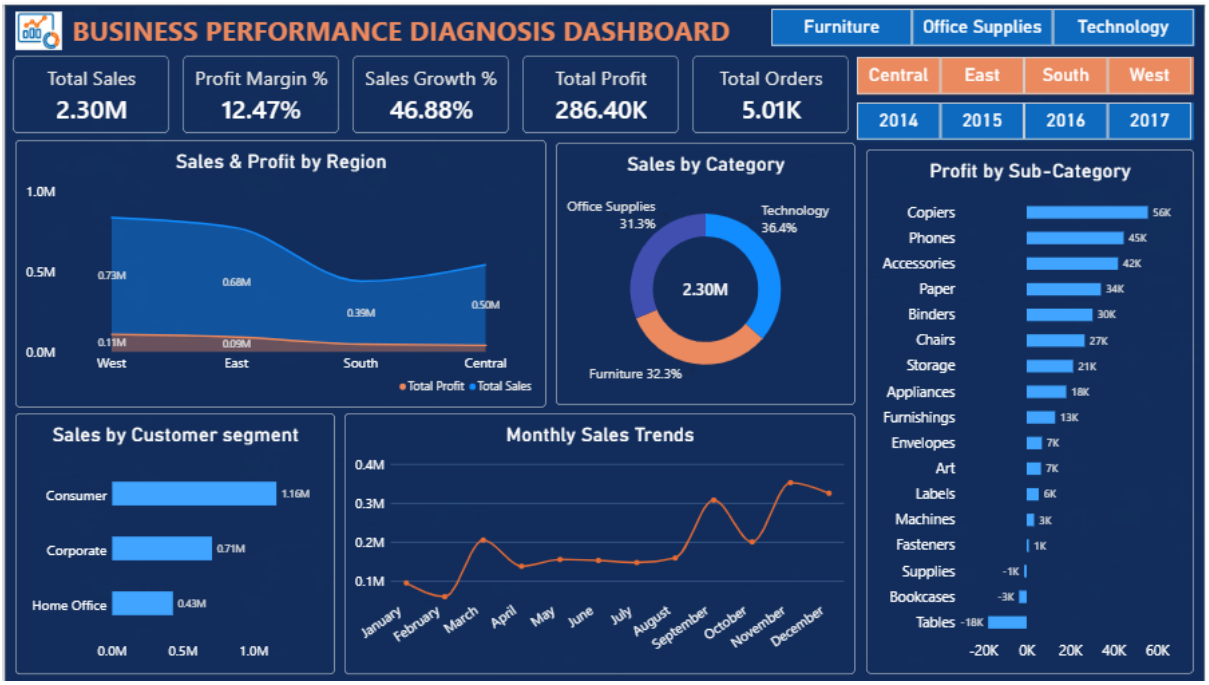
7.2 Top Loss-Making Products Breakdown

Isolating product-level profitability reveals severe capital destruction on heavily discounted high-ticket machines and conference tables:

Product Name / Identifier	Category	Sales ($)	Units	Net Loss ($)
Cubify CubeX 3D Printer Double Head (TEC-MA)	Technology	$11,099.96	9	- $8,879.97 (Worst Product)
Lexmark MX611dhe Monochrome Laser (TEC-MA)	Technology	$16,829.90	18	- $4,589.97
Cubify CubeX 3D Printer Triple Head (TEC-MA)	Technology	$7,999.98	4	- $3,839.99
Chromcraft Bull-Nose Wood Table (FUR-TA)	Furniture	$9,917.64	27	- $2,876.12
Bush Advantage Racetrack Conference (FUR-TA)	Furniture	$9,544.72	33	- $1,934.40

8. Power BI Business Performance Dashboard Architecture

To equip commercial directors and regional controllers with continuous self-service visibility, an interactive executive dashboard was developed in Microsoft Power BI Desktop featuring an elegant navy/azure corporate theme.



• **Top Executive Slicers & Filters**	Enables real-time cross-filtering across Product Categories (Furniture, Office Supplies, Technology), Geographic Regions (Central, East, South, West), and Operating Years (2014, 2015, 2016, 2017).

• **Visual Architecture & Layout**	1. Top Row: 5 High-Impact KPI Cards (Total Sales $2.30M, Profit Margin 12.47%, Sales Growth 46.88%, Total Profit $286.40K, Total Orders 5.01K) 2. Center Grid: Regional Sales & Profit Area Spline, Sales by Category Donut (Technology 36.4%, Furniture 32.3%, Office Supplies 31.3%), and Sub-Category Profit Ranked Bar Chart (highlighting Tables at -$18K) 3. Bottom Row: Customer Segment Revenue Bars (Consumer $1.16M, Corporate $0.71M, Home Office $0.43M) and Monthly Sales Trend Spline.

9. Strategic Recommendations: 3 Data-Driven Management Actions

To resolve margin erosion and immediately improve enterprise profitability, executive management should implement the following three specific, data-driven actions:

ACTION 1: CRITICAL	**1. Enforce a Strict 15% Maximum Discount Cap & Eliminate Heavy Table Markdowns** Statistical analysis proves discounts >20% cause negative margins (r = -0.22). Eliminate discounts exceeding 15% across all commercial sales, and immediately halt discounting on Furniture Tables (which lost -$17.7K). Requiring managerial sign-off for enterprise volume discounts will immediately recover an estimated $25K–$35K in annualized profit.

ACTION 2: HIGH	**2. Overhaul the Central Region's Commercial Pricing & Freight Surcharges** The Central region generates $501K in sales but yields an anemic 7.92% margin (compared to West's 14.94%), draining corporate profitability. Conduct an audit of logistics carrier rates in Texas and Illinois, enforce minimum order thresholds for heavy office furniture, and align Central pricing tiers to East/West baselines to capture ~$30K in margin lift.

ACTION 3: HIGH	**3. Rebalance Merchandising Toward High-Margin Technology & Bundled Solutions** Technology delivers 17.40% profit margin and accounts for over half of total operating profit ($145.5K). Reallocate promotional marketing toward Copiers, Phones, and Accessories, while discontinuing chronic loss-making hardware (such as Cubify 3D Printers). Bundle low-margin Furniture with high-margin Accessories to safeguard basket economics.

10. Conclusion & Profit Recovery Roadmap

This business performance diagnosis has demonstrated that the company's profit stagnation is not a market demand failure, but an internal pricing and merchandising distortion. While sales expanded by 46.88% to $2.30M, aggressive discounting eroded margins down to 12.74% in 2017.

By implementing the three recommended actions—capping discounts, fixing Central region logistics, and shifting sales mix toward Technology—the business can systematically restore operating margins above 16.0% within the next 12 months.

Strategic Performance Focus	Current Baseline (2017)	12-Month Target State
Overall Operating Profit Margin	12.47% (12.74% in 2017)	> 16.00% Across All Operations
Furniture Category Profit Margin	2.49% ($18.45K Profit)	> 8.50% via Table Price Floor
Central Region Operating Margin	7.92% ($39.71K Profit)	> 12.50% Through Freight Surcharges
Loss-Making Product Hemorrhage	-$21K+ (Tables/Machines)	Reduce Loss Drag by 75% (<$5K)
Average Transaction Discount	15.62% (Spikes to 80%)	Capped at 15.0% Max Policy

Executing this profit turnaround framework will transition the enterprise from unmonitored volume chasing to high-margin, value-accretive growth, securing sustainable corporate profitability.